YIELD ENHANCEMENT · AUTO DN

AutoDN

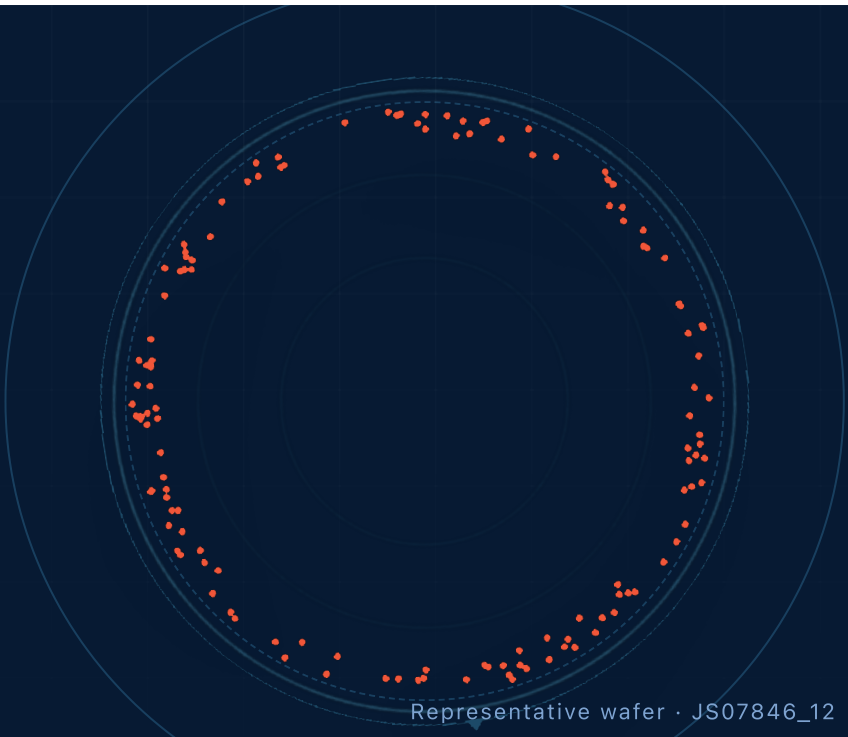
缺陷案例分析报告

从 Defect Library 匹配到 HL Step 的全链路判断，包含 OOC / OOS / Level / 证据链与处置建议。

LOT	PRODUCT	SCAN STEP
JS07846	S062A01A	MP1_ASI

DEFECT	CASE ID
Buried particle	ADN260101001

DEMO · MOCK DATA | CASE MAIN | DN YES



Representative wafer · JS07846_12

DEFECT LIBRARY	OOC WAFERS	OOS / SCRAP	HIGH LIGHT LEVEL	HL STEP
DL20260401	8 / 25	2	L2	008800
match level 10	out of control	scrap wafers	severity	DEP-01 · DEP-A

00 Executive Summary

一页看懂：结论、处置建议与证据链

结论

DN Flag YES

本卡 lot **JS07846** 在 **MP1_ASI** 站点发现 **Buried particle** 异常：**8 片 OOC**、**2 片达到 Scrap**，高亮等级 **L2**；三路证据一致指向 **008800 / DEP-01**。

3 · DRB | 5 · Scrap | 7 · Release | 8 · Release | 12 · Scrap | 15 · Release | 18 · Release | 19 · DRB

Lot Handling	NG wafer scrap
Tool Action	Down tool
Wafer List	JS07846_12

1 **DL** 匹配

2 **OOC** 失控片

3 **OOS** 报废

4 **Level** 等级

5 **HL** 高亮步骤

6 **DN** 结论

DEFECT LIBRARY

DL20260401 · level 10

精确命中 scan_step + defect_class；库条目的 Map Type 与本卡主要分布一致。

OOC 形成路径

阈值 6 片 + Special Map + IOU

基础阈值命中 6 片；Special Map 补入非 Random 分布；IOU=0.92 补入 W08。

HL 三路证据

008800 / DEP-01

Defect Library、历史 Defect Notice、Common Tool 一致指向 008800；chamber DEP-A。

01 Defect Library Match

这个 defect 在库里有先例吗?

SCAN STEP
MP1_ASI

DEFECT CLASS
Buried particle

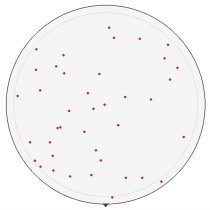
MATCH LEVEL
10

SELECTED
DL20260401

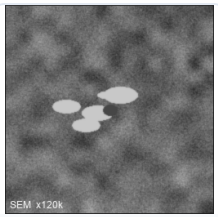
DL20260401

SELECTED

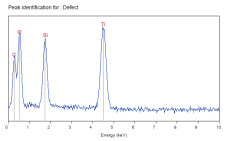
Buried particle · Random



Map



SEM



EDX

HL Spec	**Nor.Count≥ 10ea**	Scrap Spec	**D.D%≥ 10%**
Killer	**Y**	HL Step	**008800**

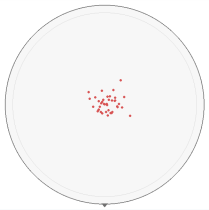
Buried particle: irregular bright blob partially embedded under the film, 100-200 nm.

图像中央可见一个不规则、部分埋入膜层的颗粒状凸起，边缘有暗色轮廓。

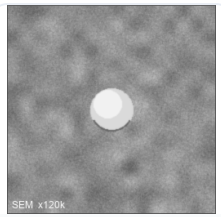
DL20260402

REJECTED

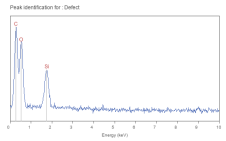
Buried particle · Center



Map



SEM



EDX

HL Spec	**自定义 暂不HL**	Scrap Spec	**自定义 暂不报废**
Killer	**N**	HL Step	**009600**

Surface particle cluster, rounded, often several particles grouped together.

图像中央可见多个大小不一的圆润颗粒聚集，呈簇状分布。

匹配逻辑

共 2 条候选。按 defect map 分布匹配度 + 是否有可执行 HL Spec + Spec 是否能被本卡 lot 触发综合评分，DL20260401 得分最高（0.936），其 map_type=Random，与当前 lot 主要 defect map 类型一致；其余候选见明细。

02 OOC · Out of Control

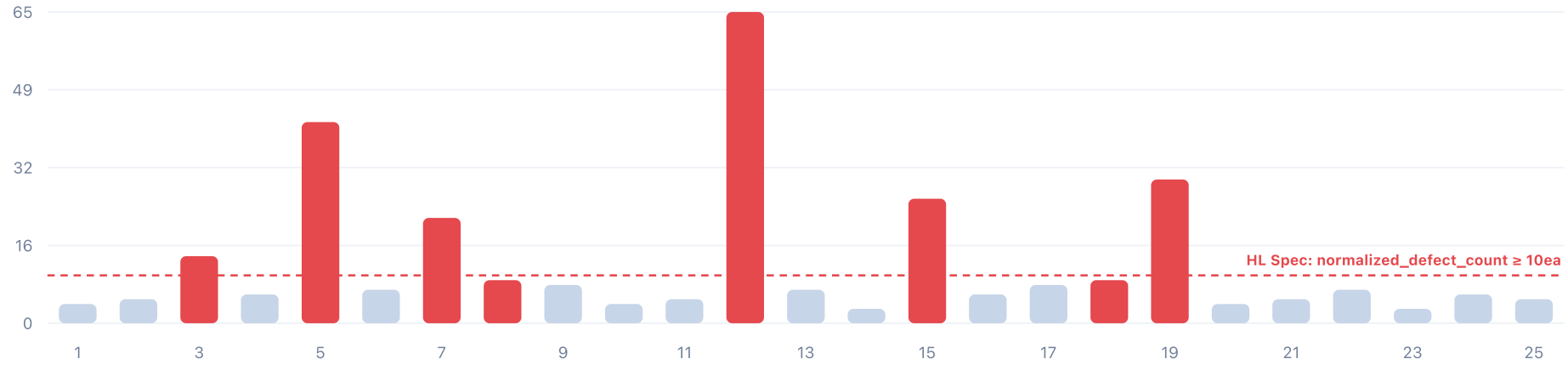
哪些 wafer 失控?

OOC	基础阈值	SPECIAL MAP	IOU 扩展
8 / 25	**6**	**+2**	**+1**

最终失控片 | HL Spec 直接命中 | 非 Random 分布补入 | 相似 map 补入

Normalized Defect Count · 逐片判定

红色 = OOC



WAFER	MAP	SUBMAP TYPE	DEFECT COUNT	NOR. DEFECT	D.D%	DIE COUNT	STATUS
JS07846_01		Random	5	4	1.5%	5	OK
JS07846_02		Random	6	5	1.8%	6	OK
JS07846_03		Center	18	14	6.0%	11	OOC
JS07846_04		Random	7	6	2.0%	7	OK

WAFER	MAP	SUBMAP TYPE	DEFECT COUNT	NOR. DEFECT	D.D%	DIE COUNT	STATUS
JS07846_05		Edge Ring	96	42	18.0%	62	OOC
JS07846_06		Random	8	7	2.2%	8	OK
JS07846_07		Center	40	22	4.0%	17	OOC
JS07846_08		Edge Ring	120	9	3.0%	76	OOC
JS07846_09		Random	9	8	2.4%	9	OK
JS07846_10		Random	5	4	1.5%	5	OK
JS07846_11		Random	6	5	1.8%	6	OK
JS07846_12		Edge Ring	130	65	22.0%	81	OOC
JS07846_13		Random	8	7	2.2%	8	OK
JS07846_14		Random	4	3	1.2%	4	OK

WAFER	MAP	SUBMAP TYPE	DEFECT COUNT	NOR. DEFECT	D.D%	DIE COUNT	STATUS
JS07846_15		Center	30	26	5.0%	14	OOC
JS07846_16		Random	7	6	2.0%	7	OK
JS07846_17		Random	9	8	2.4%	9	OK
JS07846_18		Center	40	9	2.7%	13	OOC
JS07846_19		Local	80	30	7.0%	11	OOC
JS07846_20		Random	5	4	1.5%	5	OK
JS07846_21		Random	6	5	1.8%	6	OK
JS07846_22		Random	8	7	2.2%	8	OK
JS07846_23		Random	4	3	1.2%	4	OK
JS07846_24		Random	7	6	2.0%	7	OK

WAFER	MAP	SUBMAP TYPE	DEFECT COUNT	NOR. DEFECT	D.D%	DIE COUNT	STATUS
JS07846_25		Random	6	5	1.8%	6	OK

Special Map OOC

库条目的 map_type 为 `Random`，但下列 wafer 的实际 submap 为非 Random，按规则补入：

3 5 7 8 12 15 18 19

其中 ['8', '18'] 不在基础 OOC 中。

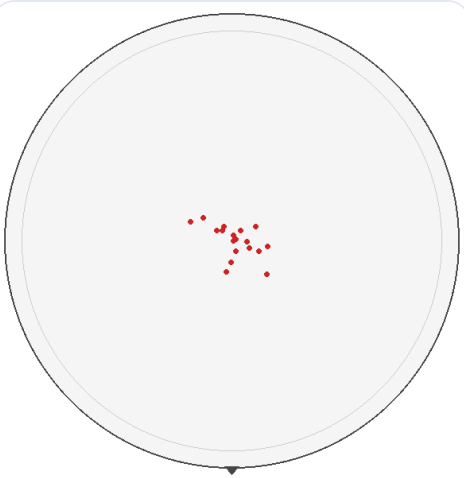
IOU 相似 Map

以最严重 OOC wafer 为基准，重叠面积 / 基准面积 > 0.7 的未 review wafer 补入：

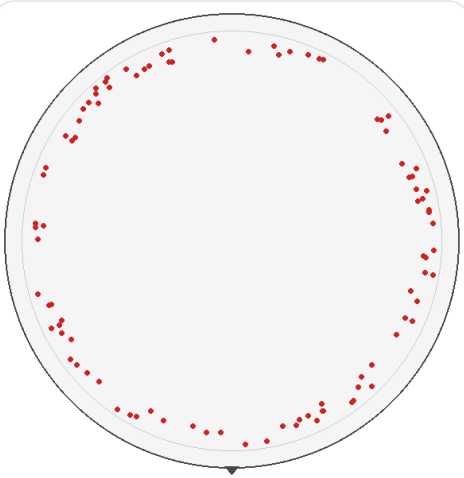
8

JS07846_08	0.92
JS07846_05	0.01
JS07846_01	0.00
JS07846_02	0.00
JS07846_03	0.00

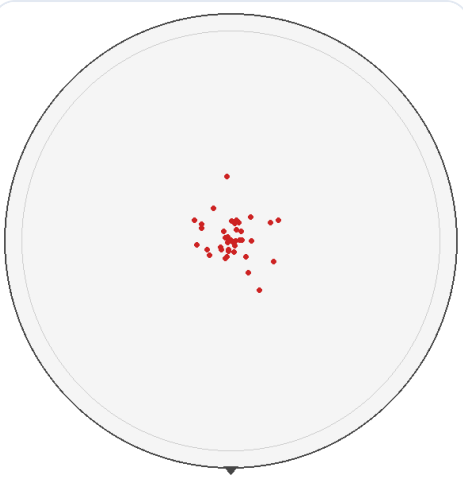
OOC wafer maps



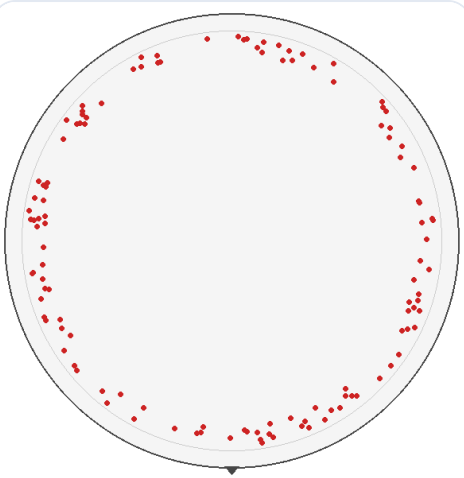
JS07846_03
Center · 18 defects



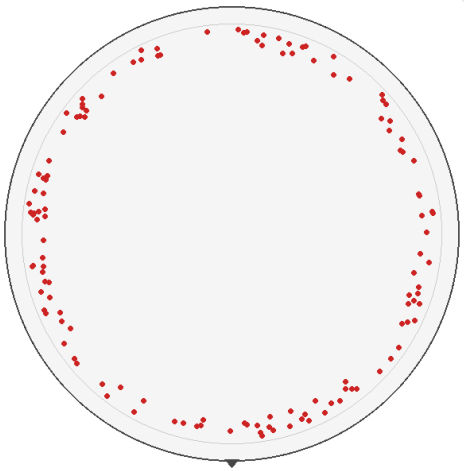
JS07846_05
Edge Ring · 96 defects



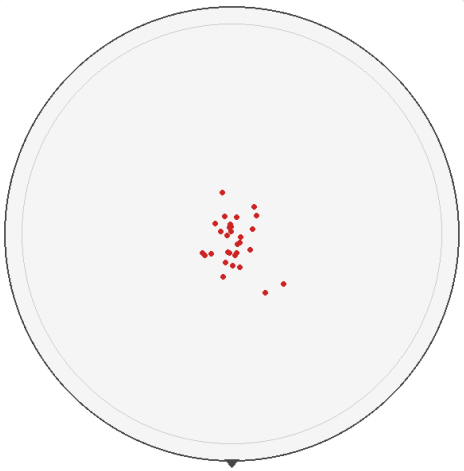
JS07846_07
Center · 40 defects



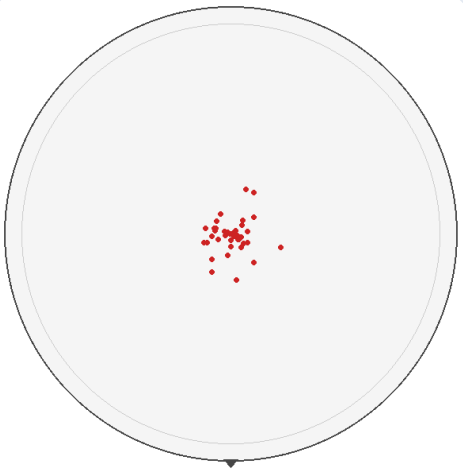
JS07846_08
Edge Ring · 120 defects



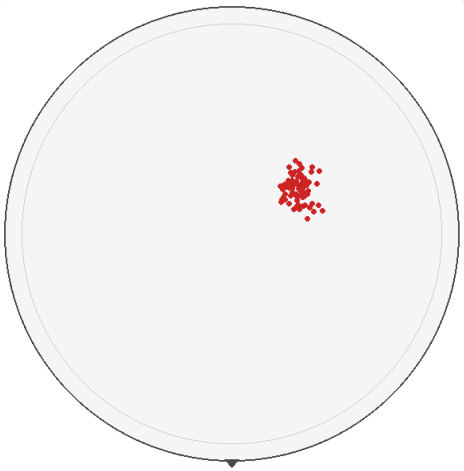
JS07846_12
Edge Ring · 130 defects



JS07846_15
Center · 30 defects



JS07846_18
Center · 40 defects



JS07846_19
Local · 80 defects

03 OOS · Out of Spec

OOC 里哪些 wafer 要报废?

OOS / SCRAP

2

达到 Scrap Spec

SCRAP SPEC

D.D% ≥ 10%

仅对 OOC 判定

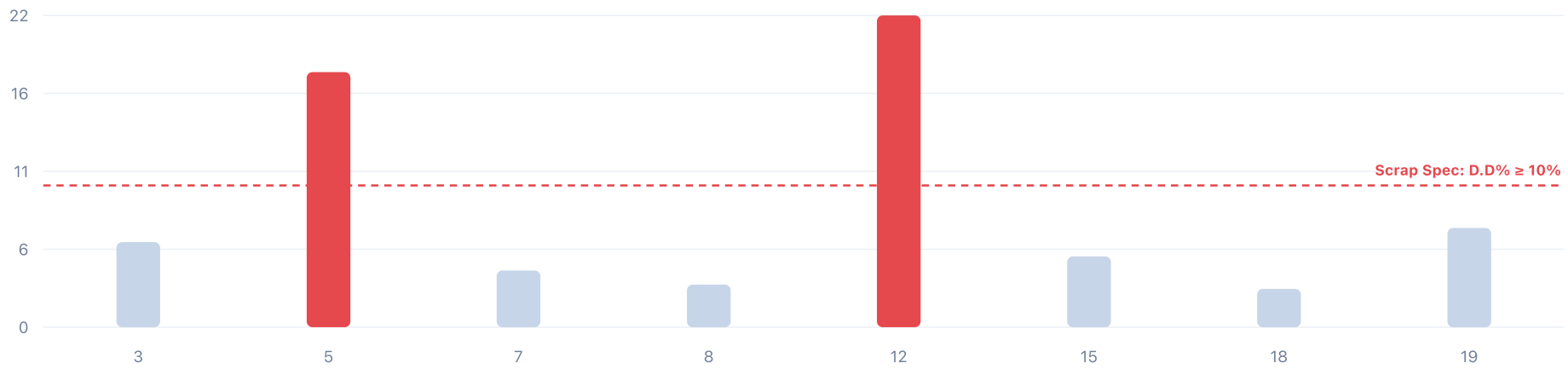
KILLER

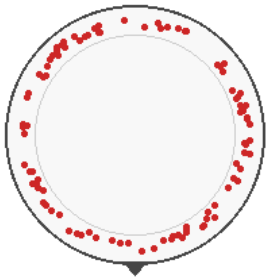
Y

killer defect

D.D% · OOC wafers

红色 = Scrap



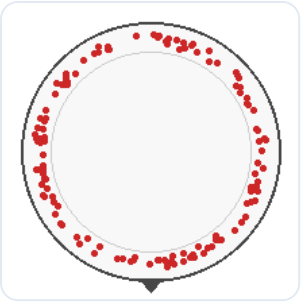


JS07846_05

SCRAP

D.D%	NOR. DEFECT	DEFECT COUNT
18.0%	42	96

Edge Ring map · killer defect



JS07846_12

SCRAP

D.D%	NOR. DEFECT	DEFECT COUNT
22.0%	65	130

Edge Ring map · killer defect

判定说明

从 DL20260401 提取 OOS 标准：Normalized Defective Die Percentage >= 10%，仅对 OOC wafer 判定；满足条件：['5', '12']。

04

Level · High Light Level

这个 case 几级?

HIGH LIGHT LEVEL

L2

8 OOC · 2 OOS

OOC = 8pcs → 满足 Level 2（>=2pcs OOC）；OOS = 2pcs 且 killer_defect=Y → 满足 Level 2（1~2pcs OOS killer）。

L1	L2	L3	L4
蓝色	**黄色**	**橙色**	**红色**
1pcs OOC	≥2pcs OOC / 1~2pcs OOS(killer) / 刮伤 =1	≥12pcs OOC / ≥3pcs OOS(killer) / 刮伤 ≥2	≥10pcs scrap(killer) / 重复 L3
关注 / 复查	Hold & 通知	立即 Hold & 升级	重大异常 / 上报

触发 LEVEL	规则
2	OOC = 8pcs → 满足 Level 2（>=2pcs OOC）
2	OOS = 2pcs 且 killer_defect=Y → 满足 Level 2（1~2pcs OOS killer）

为什么不是更高等级?

OOC = 8pcs，未达到 L3 的 ≥12pcs；OOS = 2pcs，未达到 L3 的 ≥3pcs（killer）。机械刮伤/RE defect 条件未触发，因此取最高命中 L2。

---

05 Similar Defect Notice

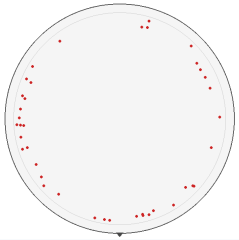
历史上有相似案例吗?

按 defect type / map type / level / HL step 四个维度评分，保留 score ≥ 0.5 的 Top N 历史 DN，作为 HL Step 的辅助证据。

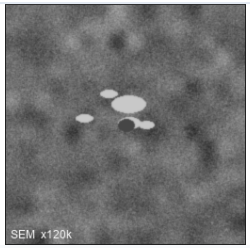
DN20251228005

Buried particle · Edge Ring

0.90 similarity



Map



SEM

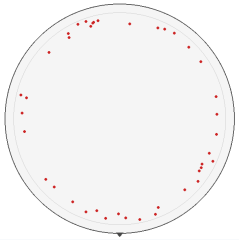
Rule Level **L2**

HL Step **008800**

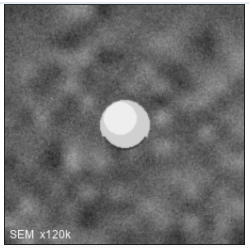
DN20251102007

Particle · Edge Ring

0.50 similarity



Map



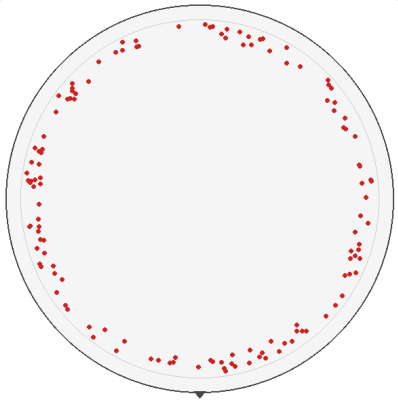
SEM

Rule Level **L3**

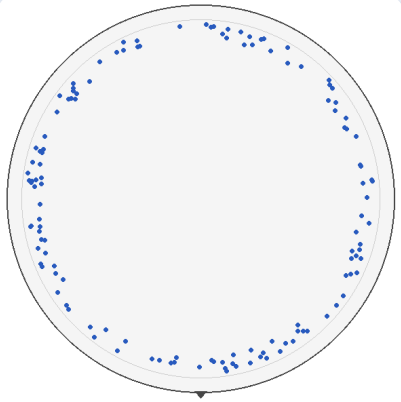
HL Step **008800**

06 HL Step · High Light Step

问题最可能出在哪一道工艺?



Total Map



Target Defect Map

REPRESENTATIVE NG WAFER

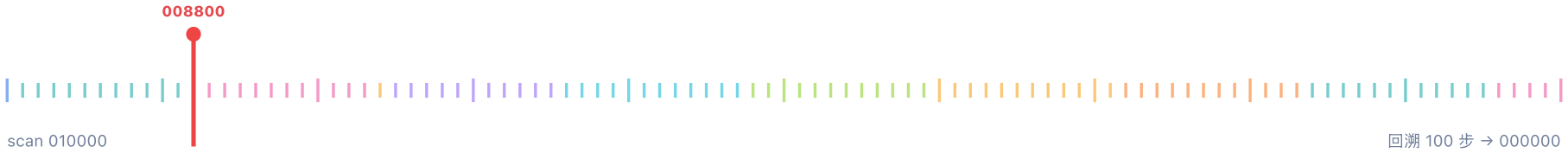
JS07846_12

DEFECT COUNT	SUBMAP TYPE	NOR. DEFECT	D.D%
130	Edge Ring	65	22.0%

选 OOC 中 ndefect 最大的一片，在过货记录中定位扫描 recipe，向前回溯最多 100 步。

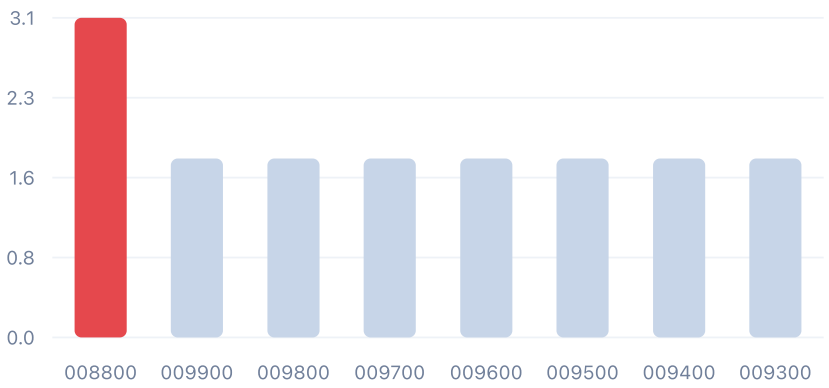
Wafer History · 100-step backtrack

红色 = HL 候选步骤

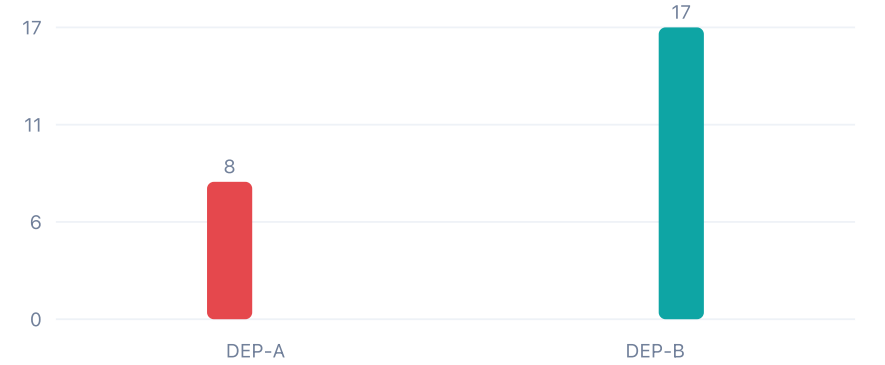


Common Tool · Top 8

Common Score



Chamber split · 008800



01

Defect Library

hl_step = 008800，模块 DEP

02

历史 Defect Notice

相似 DN 的 HL step 指向 ['008800']

03

Common Tool

最高分步骤 ['008800', '009900', '009800']

008800

tool · DEP-01

DEP-A

score 1

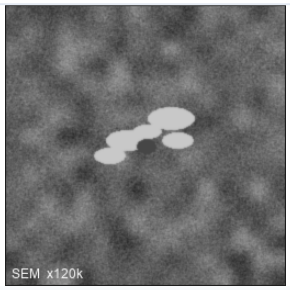
DN 结论 / 处置

DN Flag	YES
Lot Handling	**NG wafer scrap**
Tool Action	**Down tool**
Wafer List	**JS07846_12**

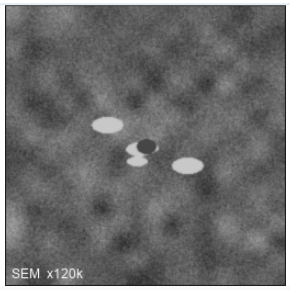
合并推理

选取最严重 NG wafer；在其过货记录中定位 010000，向前回溯 100 步； Defect Library 指向 008800；历史 DN 指向 ['008800']； Common Tool 最高分步骤 ['008800', '009900', '009800']；合并后保留 ['008800']。

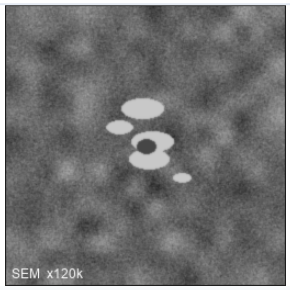
- **008800** Defect Library DL20260401 指向 step 008800（DEP）
- **008800** 历史 Defect Notice ['DN20251228005', 'DN20251102007'] 的 high light step 同为 008800
- **008800** Common Tool: score=3.125, bad_ratio=1.0, defect_ratio=1.0, top chamber=DEP-A



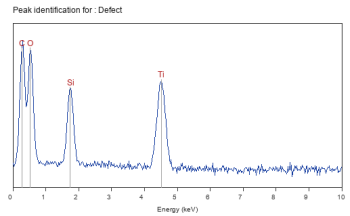
图像中央可见一个不规则、部分埋入膜层的颗粒状凸起，边缘带暗色轮廓



图像中央可见一个不规则、部分埋入膜层的颗粒状凸起，边缘带暗色轮廓



图像中央可见一个不规则、部分埋入膜层的颗粒状凸起，边缘带暗色轮廓



EDX

▶ **展开完整过货记录（101 步）**

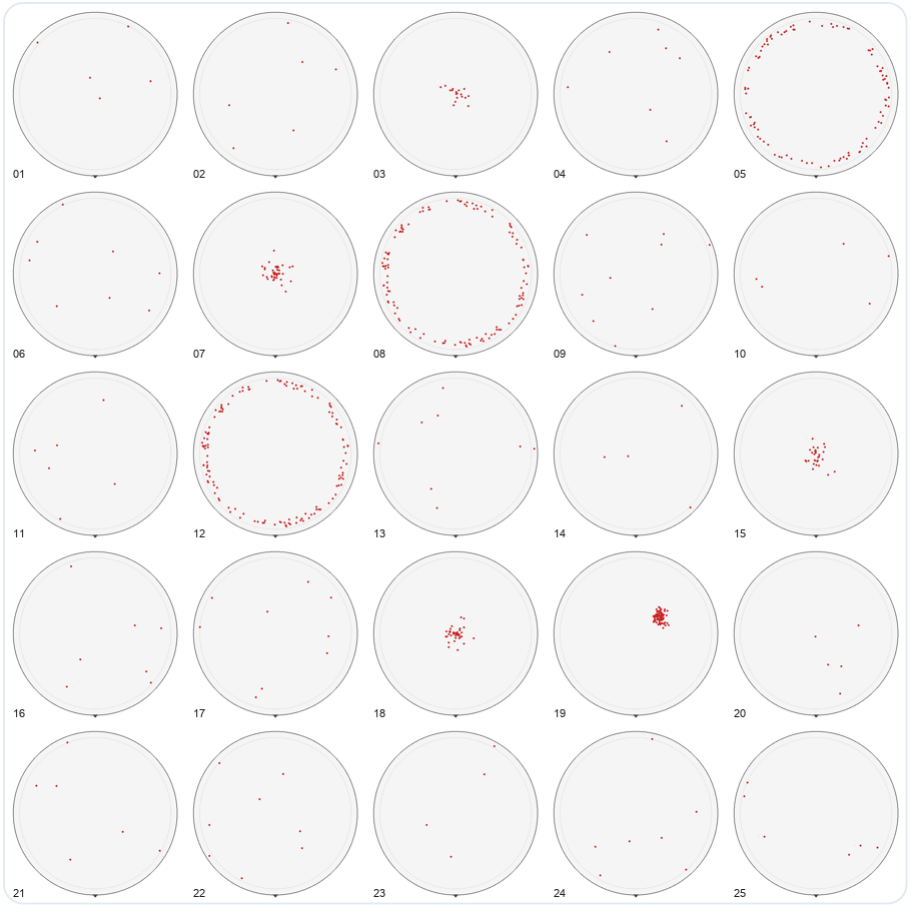
07 Summary · DN Result

最终结论、全片地图与可回写系统的 JSON

```
L2+JS07846+S062A01A+MP1_ASI+Buried particle+ADN260101001
```

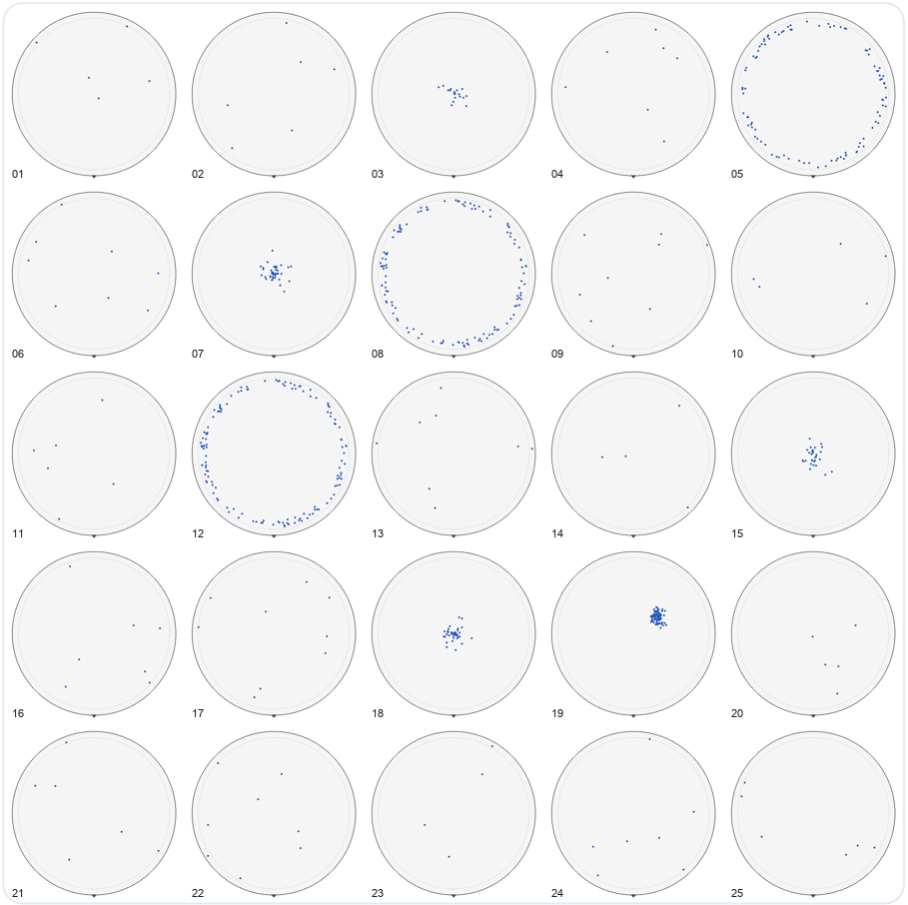
Total Map · 25 wafers

全缺陷

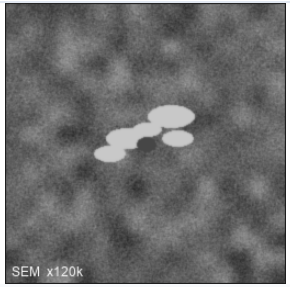


Target Defect Map · 25 wafers

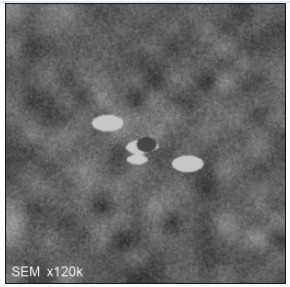
目标 defect class



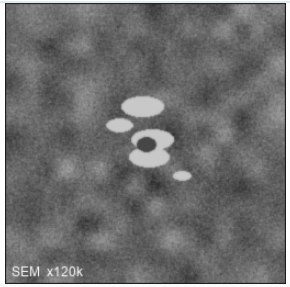
Representative wafer · JS07846_12



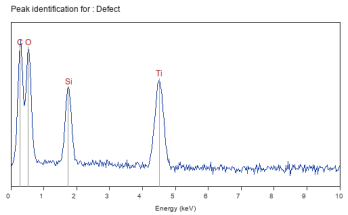
图像中央可见一个不规则、部分埋入膜层的颗粒状凸起，边缘带暗色轮廓



图像中央可见一个不规则、部分埋入膜层的颗粒状凸起，边缘带暗色轮廓



图像中央可见一个不规则、部分埋入膜层的颗粒状凸起，边缘带暗色轮廓



EDX

Wafer handling

WAFER	OOC	OOS	D.D%	HANDLING
JS07846_03	OOC		6.0%	DRB
JS07846_05	OOC	SCRAP	18.0%	Scrap
JS07846_07	OOC		4.0%	Release
JS07846_08	OOC		3.0%	Release
JS07846_12	OOC	SCRAP	22.0%	Scrap
JS07846_15	OOC		5.0%	Release
JS07846_18	OOC		2.7%	Release
JS07846_19	OOC		7.0%	DRB

final_result.json


```
{
  "type": "AutoDN",
  "autoDnId": "ADN260101001",
  "dnFlag": "yes",
  "defectLibraryId": "DL20260401",
  "defectLibraryLevel": "10",
  "ngWaferList": "3,5,7,8,12,15,18,19",
  "waferHandleList": "DRB,Scrap,Release,Release,Scrap,Release,Release,DRB",
  "defectLevel": "L2",
  "highLightStepList": [
    {
      "step_number": "008800",
      "tool_type": "tool",
      "tool": "DEP-01",
      "chambers": [
        "DEP-A"
      ]
    }
  ],
  "hlInfo": [
    {
      "hlStep": "008800",
      "toolType": "tool",
      "tool": "DEP-01",
      "chambers": [
        "DEP-A"
      ]
    }
  ]
```


08 Appendix · Data & Verification

数据来源、自检与边界说明

数据与插件

- **数据**：AutoDN mock case（25 片 wafer；DMS 三表、缺陷坐标、Defect Library、历史 DN、过货记录）。
- **图片**：Map/SubMap 由缺陷坐标绘制；SEM/EDX 为合成图，mock VLM 输出描述。
- **判断**：当前报告为 `auto` 规则替身；`agent` 模式可由当前模型填写 decisions。
- **真实数据**：Data Studio 数据集或 MCP/外部服务；需要图片时走 full pipeline。

端到端自检

检查项	实际	期望	结果
retrieval_level	"10"	"10"	PASS
defect_library_id	"DL20260401"	"DL20260401"	PASS
ooc	["3", "5", "7", "8", "12", "15", "18", "19"]	["3", "5", "7", "8", "12", "15", "18", "19"]	PASS
oos	["5", "12"]	["5", "12"]	PASS
level	"2"	"2"	PASS
hl_step	"008800"	"008800"	PASS

说明

本报告用于 Demo 演示与流程验收。所有数字来自脚本输出；mock 场景的结论与 ground truth 一致，不代表真实产线结论。